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PROJECT

Urban Life in Motion

Supervised By

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Submitted by

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Group Member Information

Member Name	ID	Assigned Area
Adham Samin	23-53329-3	Sky, clouds, rain, rain clouds, sun, moon, welcome page, and related transitions
Topu Roy	23-53318-3	Road, cars, traffic light, and traffic transitions
Al Rahit	23-53316-3	Buildings, building-related scene details, wind wave transition
Utsho Karmakar	23-53327-3	Ground, trees, lamp posts, flags, and wind wave transition

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1. INTRODUCTION

The project “Urban Life in Motion” is a 2D computer graphics animation project developed using OpenGL and GLUT. The main objective of this project is to represent a modern urban environment with moving and interactive elements. The scene includes buildings, roads, cars, traffic lights, trees, flags, lamp posts, sky, clouds, sun, moon, rain, and weather transitions.

The project focuses on applying basic computer graphics concepts such as drawing primitives, object composition, animation, keyboard interaction, color variation, and object transformation. Different objects are created using OpenGL primitives like GL_QUADS, GL_TRIANGLES, GL_POLYGON, GL_LINES, and GL_POINTS.

The project also includes interactive controls. The user can start the project from a welcome screen, control the traffic light, switch between day and night, turn rain on or off, and control wind direction. The project code uses separate drawing functions, global variables, update functions, and keyboard functions, following the faculty-provided OpenGL structure.

2. Proposal

The proposed project is an animated city environment that shows urban life with both environmental and traffic-based activities. The city scene contains roads, cars, buildings, greenery, weather effects, and interactive controls.

The main idea is to create a lively urban environment where:

- Cars move on the road.
- Traffic lights control car movement.
- The sky changes between day and night.
- Rain can be turned on or off.
- Clouds and rain clouds appear in the sky.
- Wind affects trees, flags, and cloud movement.
- A welcome page appears before the main animation starts.

The project demonstrates how simple OpenGL drawing techniques can be combined to create a complete animated scene.

3. SCHEMATIC DIAGRAM

Figure 01: Welcome Page

Figure 02: Project main Interface

4. List of Objects

The project includes the following objects:

1. Welcome / Start screen
2. Project title text
3. Group member information text
4. Animation control instruction text
5. Sky
6. Stars
7. Sun
8. Sun rays
9. Moon
10. Clouds
11. Rain clouds
12. Rain drops
13. Wind effect lines
14. Buildings
15. Hospital building
16. School building
17. Fire service station
18. Police station
19. Building doors
20. Building windows
21. Building roofs
22. Ground / grass area
23. Trees
24. Tree trunks
25. Tree leaves
26. Lamp posts
27. Lamp lights
28. Bangladesh flags
29. Flag poles
30. Road
31. Footpath
32. Road borders
33. Middle divider
34. Lane markers
35. Zebra crossing
36. Traffic light pole
37. Traffic light box
38. Red traffic signal
39. Yellow traffic signal
40. Green traffic signal
41. Cars
42. Car bodies
43. Car cabins
44. Car windows
45. Car headlights
46. Car tail lights
47. Car wheels

5. Functions to Represent the Objects

These are the main functions used to draw or represent the objects in the project:

Function Name	Object / Purpose
drawText()	Draws text on the screen
drawStartScreen()	Draws the welcome/start screen
drawSky()	Draws the sky background and gradient
drawStars()	Draws stars during night
drawSunMoon()	Draws the sun, sun rays, and moon
drawClouds()	Draws moving clouds
drawRainClouds()	Draws dark rain clouds
drawRain()	Draws rain drops
drawWindEffect()	Draws wind direction lines
drawBuildings()	Draws all buildings and building details
drawGround()	Draws the grass/ground area
drawWindTree()	Draws a single wind-affected tree
drawTrees()	Draws all trees
drawLampPosts()	Draws lamp posts and lights
drawFlags()	Draws Bangladesh flags
drawRoad()	Draws road, footpath, divider, lane markers, and zebra crossing
drawTrafficLight()	Draws traffic light pole, box, and signal lights
drawSingleCar()	Draws one car
drawCars()	Draws all cars
drawCircle()	Helper function for circular objects
drawZebraLane()	Helper function for zebra crossing lanes

6. Interactive Functions

Interactive functions are used to control animation through keyboard input.

Main Interactive Function

void handleKeypress(unsigned char key, int x, int y)

This function handles all keyboard controls.

Keyboard Controls

Key	Action
S	Starts the project from the welcome screen
R	Changes traffic light to red
Y	Changes traffic light to yellow
G	Changes traffic light to green
Enter	Pauses or resumes car movement
D	Activates day mode
N	Activates night mode
T	Turns rain on or off
L	Wind blows from left to right
K	Wind blows from right to left
O	Stops wind

7. Iterative Functions and Transitions

Iterative functions are used to update object positions and transition values repeatedly. In this project, the main iterative function is:

void update(int value)

This function is repeatedly called using:

glutTimerFunc(20, update, 0);

Transitions handled in update()

Transition / Animation	Description
Cloud movement	Clouds move continuously across the sky
Wind-based cloud movement	Clouds move faster depending on wind direction
Rain transition	Rain intensity increases or decreases smoothly
Rain drop movement	Rain drops fall downward and reset after leaving the screen
Day/night transition	Sky changes gradually between day and night
Car movement	Cars move on road lanes
Traffic transition	Cars respond to red, yellow, and green traffic signals
Wind wave transition	Trees and flags bend/wave based on wind direction

8. Task Assignment and Codes of Functions

1. Adham Samin

AssignedTask:
Sky, clouds, rain, rain clouds, sun, moon, first welcome page, and related transitions.

Functions written / used for assigned task:

- void drawText(float x, float y, const char *text);
- void drawStartScreen();
- void initRain();
- void startRainSound();
- void stopRainSound();
- void drawSky();
- void drawStars();
- void drawSunMoon();
- void drawClouds();
- void drawRainClouds();
- void drawRain();

Transition / interaction code related to task:

- **update()**
- **handleKeypress()**

Specific contribution:

- Created the welcome/start page.
- Displayed project information and controls.
- Created the sky background.
- Created sun and moon objects.
- Created cloud objects.
- Created rain cloud system.
- Created rain drop animation.
- Added rain sound.
- Added rain on/off control.
- Added smooth rain transition.
- Added cloud speed change with wind.
- Added day/night environment support.

2. Topu Roy

Assigned Task:
Road, cars, traffic light, and traffic transitions.

Functions written / used for assigned task:

- drawCircle();
- drawRoad();
- drawZebraLane();
- drawTrafficLight();
- drawSingleCar();
- drawCars();
- moveRightLane();
- moveLeftLane();
- getGreenOrYellowSpeed();

Transition / interaction code related to task:

- update()
- handleKeypress()

Specific contribution:

- Created road and footpath.
- Created road divider and lane markings.
- Created zebra crossing.
- Created traffic light.
- Created cars.
- Added car movement.
- Added traffic signal control using red, yellow, and green keys.
- Added car stopping logic at red light.
- Added slow movement during yellow light.
- Added pause/resume control for cars.

3. Utsho Karmakar

Assigned Task:
Ground, trees, lamp posts, flags, and wave transition.

Functions written / used for assigned task:

- **drawGround()**
- **drawWindTree()**
- **drawTrees()**
- **drawLampPosts()**
- **drawFlags()**
- **drawWindEffect()**
- **getWindBend()**

Transition / interaction code related to task:

- **update()**
- **handleKeypress()**

Specific contribution:

- Created ground/grass area.
- Created trees.
- Created lamp posts and lights.
- Created Bangladesh flags.
- Added wind-based wave transition for trees.
- Added wind-based wave transition for flags.
- Added wind direction control.
- Added wind visual effect lines.

4. Al Rahit

Assigned Task:
All buildings, building-related details and wave transitions

Functions written / used for assigned task:

- drawBuildings()
- drawWindEffect()
- getWindBend()

Transition / interaction code related to task:

- **update()**
- **handleKeypress()**

Specific contribution:

- Created all background buildings.
- Created all foreground buildings.
- Created hospital building.
- Created school building.
- Created fire service station.
- Created police station.
- Created building roofs.
- Created building doors.
- Created building windows.
- Added window lighting effect based on rain/night condition.
- Added wind direction control.
- Added wind visual effect lines.

9. Conclusion

The project “Urban Life in Motion” successfully presents a dynamic 2D urban environment using OpenGL and GLUT. In this project, we created a complete city scene with buildings, roads, moving cars, traffic lights, trees, flags, lamp posts, sky, sun, moon, clouds, rain, and wind effects. The scene was made interactive so that users can control traffic signals, switch between day and night, turn rain on or off, control wind direction, and start the project from a welcome screen.

Through this project, we achieved a better understanding of basic computer graphics concepts such as object drawing, primitive shapes, color handling, animation, keyboard interaction, and smooth transitions. We also learned how different visual elements can be combined together to create a meaningful animated environment. The use of timer functions helped us animate objects such as cars, clouds, rain, trees, and flags in a continuous way.

The project also helped us understand how real-life urban activities can be represented through simple graphical objects. For example, the traffic light controls the cars, rain changes the environment, wind affects the trees and flags, and day/night modes change the overall atmosphere of the city.

In the future, more features could be added to improve the project further. Pedestrians, birds, street shops, moving public transport, emergency vehicles, lightning effects, road signals, sound effects for traffic, and more realistic building designs could be included. The weather system could also be improved with thunder, fog, or snowfall. Overall, this project gave us practical experience in creating an interactive animated city scene using computer graphics